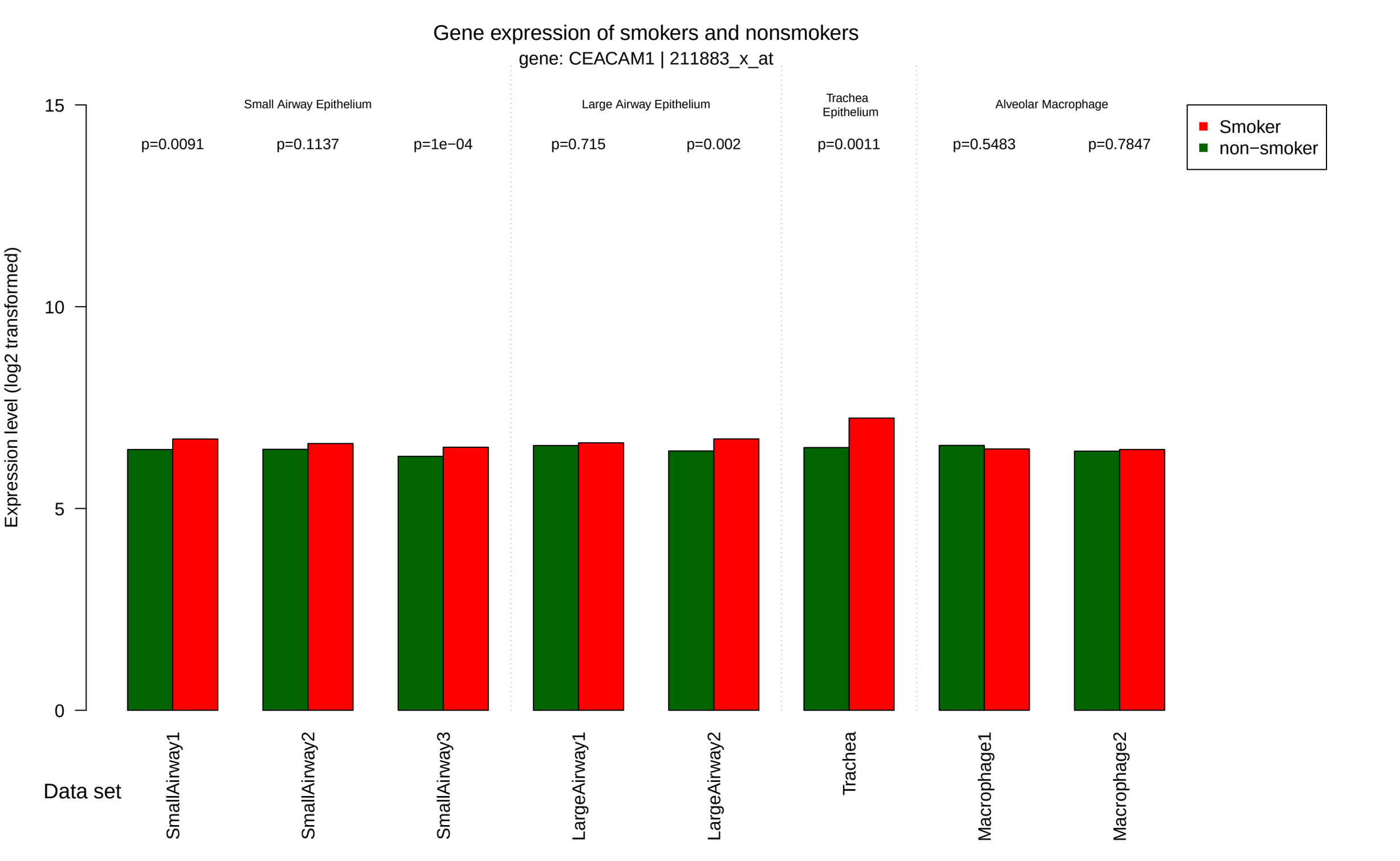
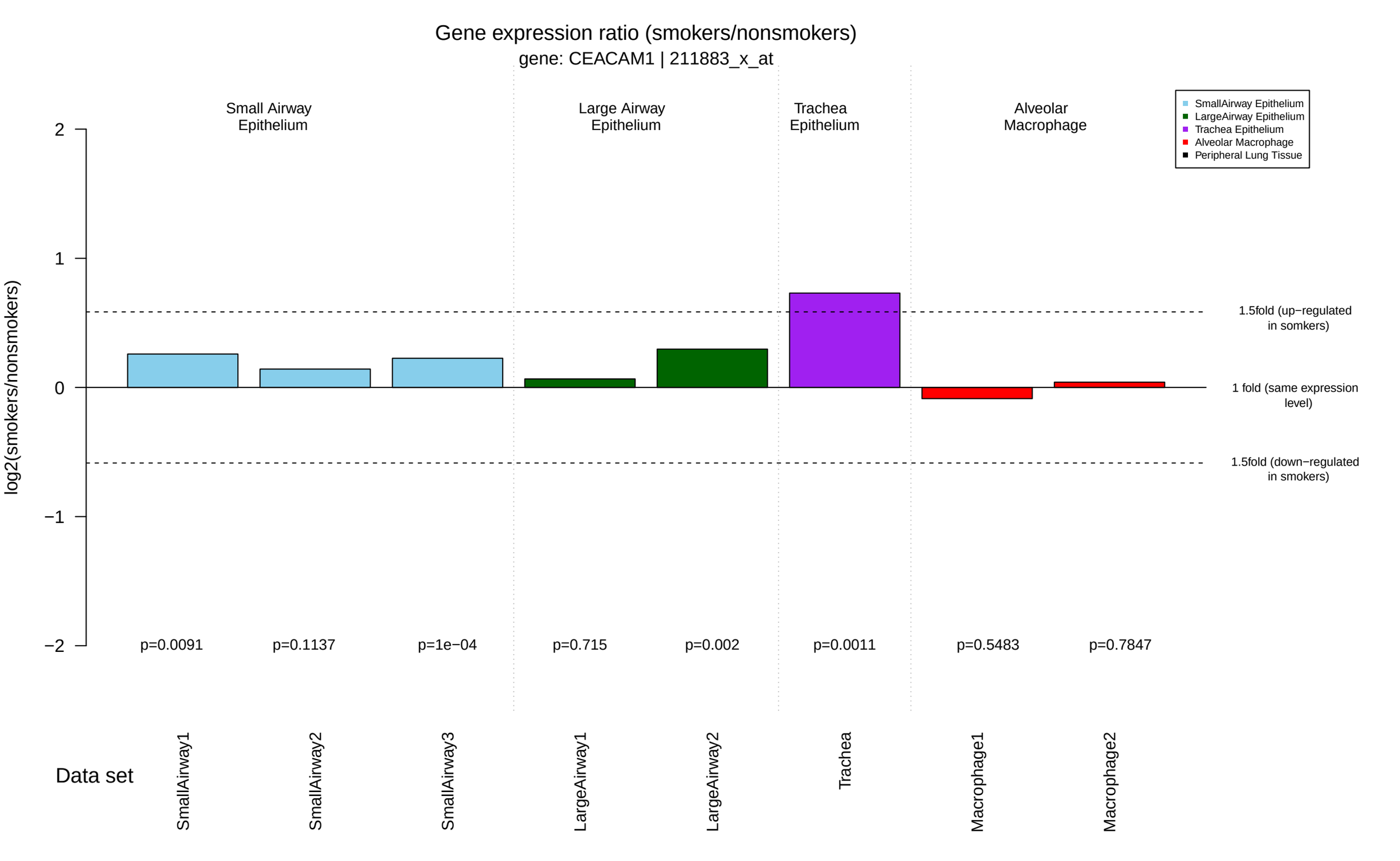
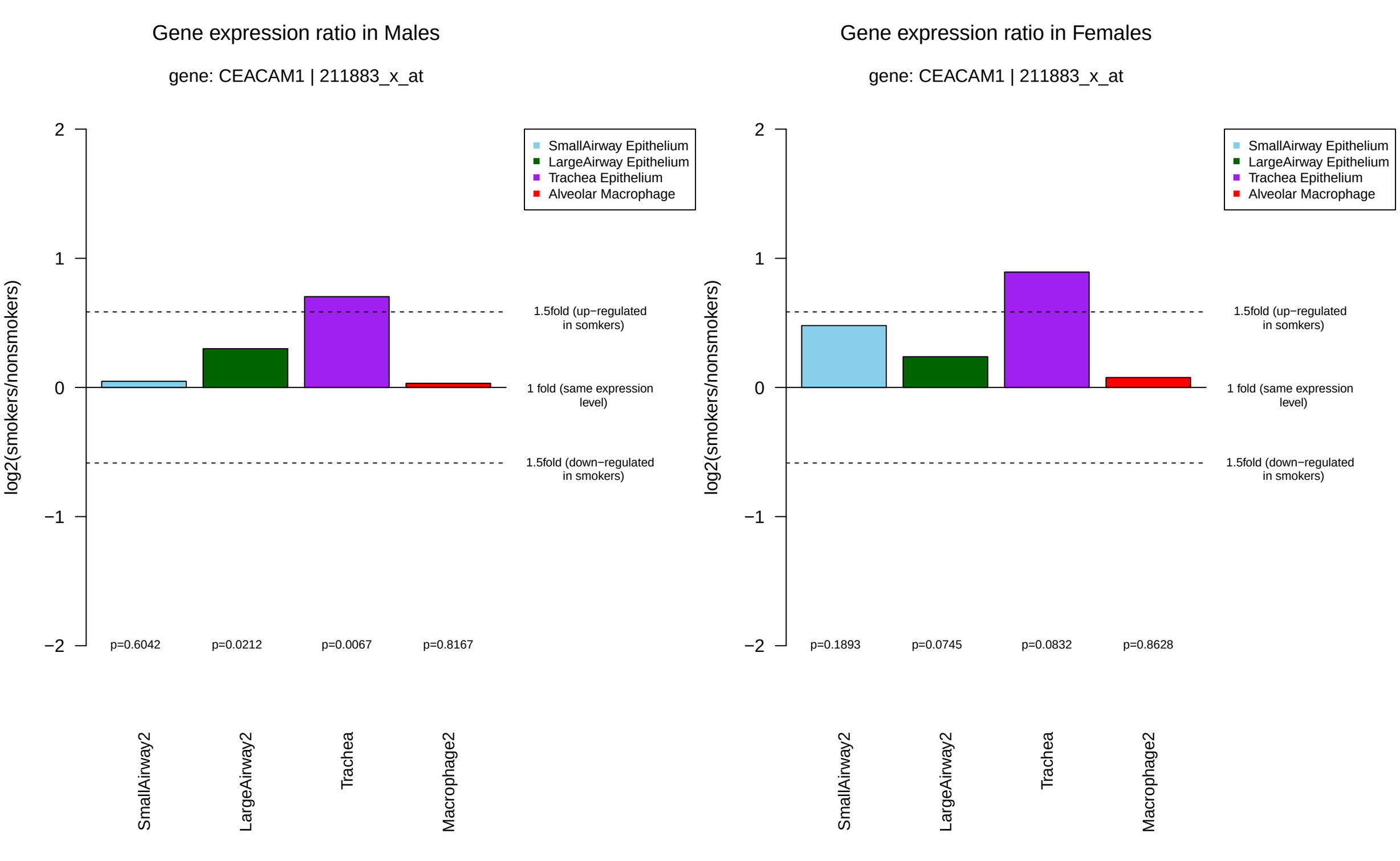
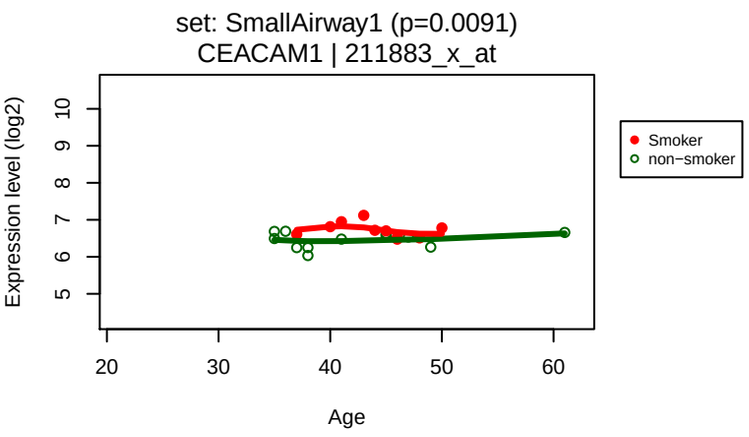
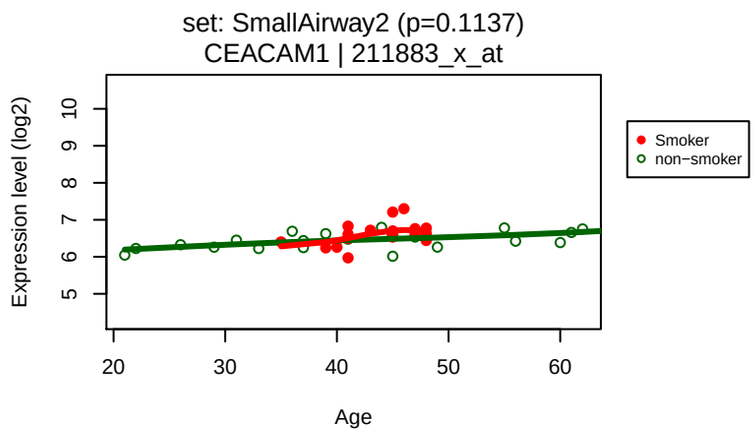
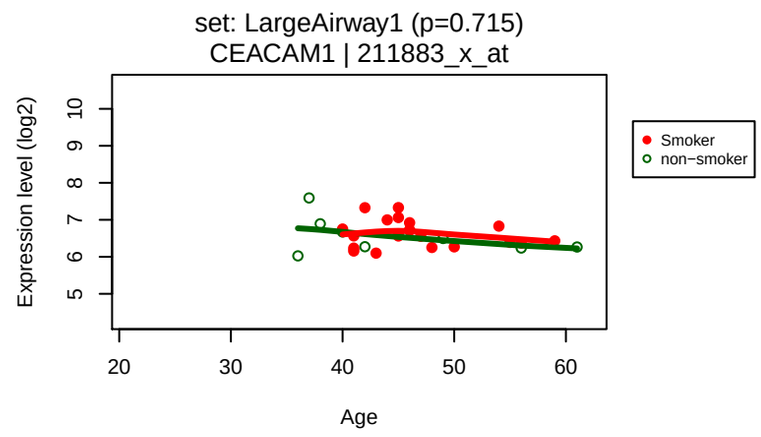

Gene expression of smokers and nonsmokers
gene: CEACAM1 | 211883_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0091
p=0.1137
p=1e−04
p=0.715
p=0.002
p=0.0011
p=0.5483
p=0.7847
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

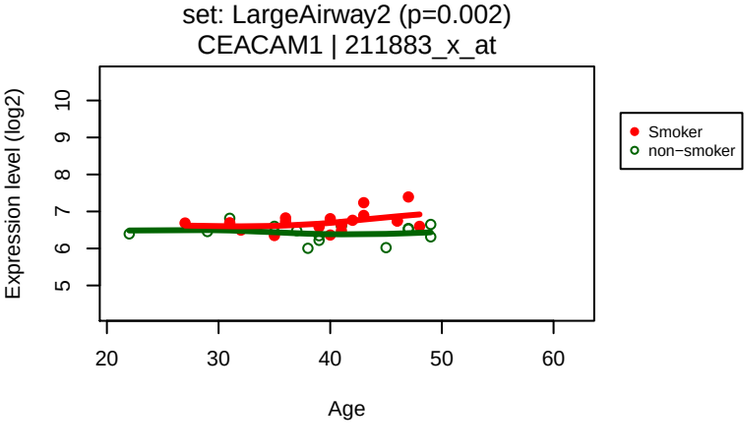

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM1 | 211883_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0091
p=0.1137
p=1e−04
p=0.715
p=0.002
p=0.0011
p=0.5483
p=0.7847
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

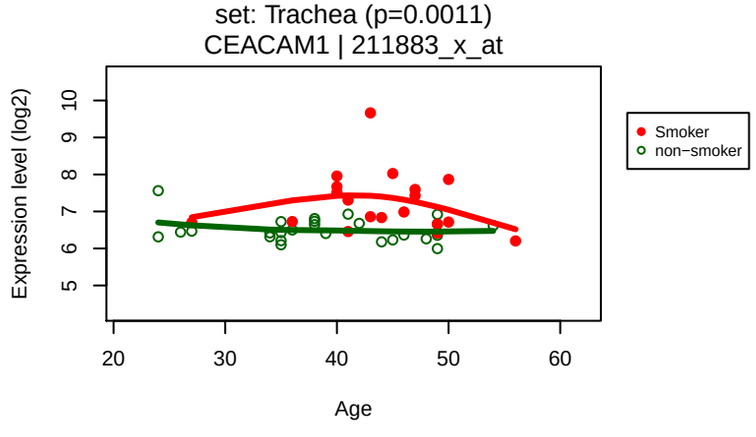

Gene expression ratio in Males
gene: CEACAM1 | 211883_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6042
p=0.0212
p=0.0067
p=0.8167
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM1 | 211883_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1893
p=0.0745
p=0.0832
p=0.8628
SmallAirway2
LargeAirway2
Trachea
Macrophage2

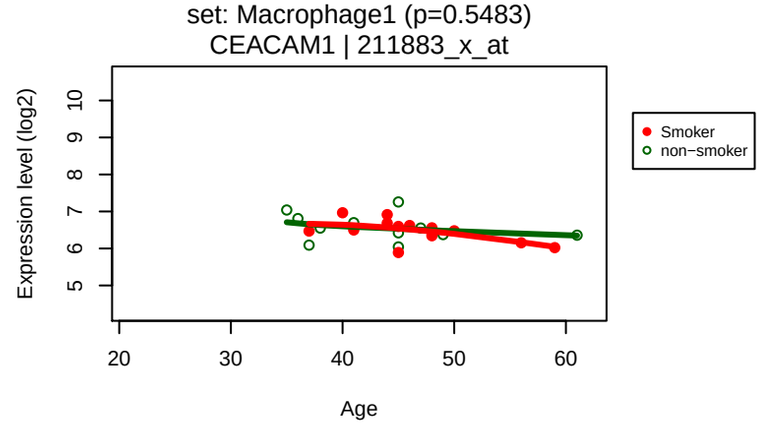

set: SmallAirway1 (p=0.0091)
CEACAM1 | 211883_x_at
Expression level (log2)
Age
Smoker
non−smoker

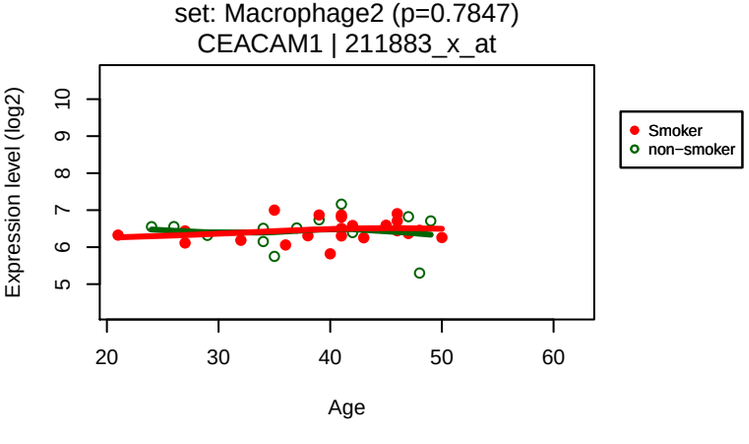

set: SmallAirway2 (p=0.1137)
CEACAM1 | 211883_x_at
Expression level (log2)
Age
Smoker
non−smoker

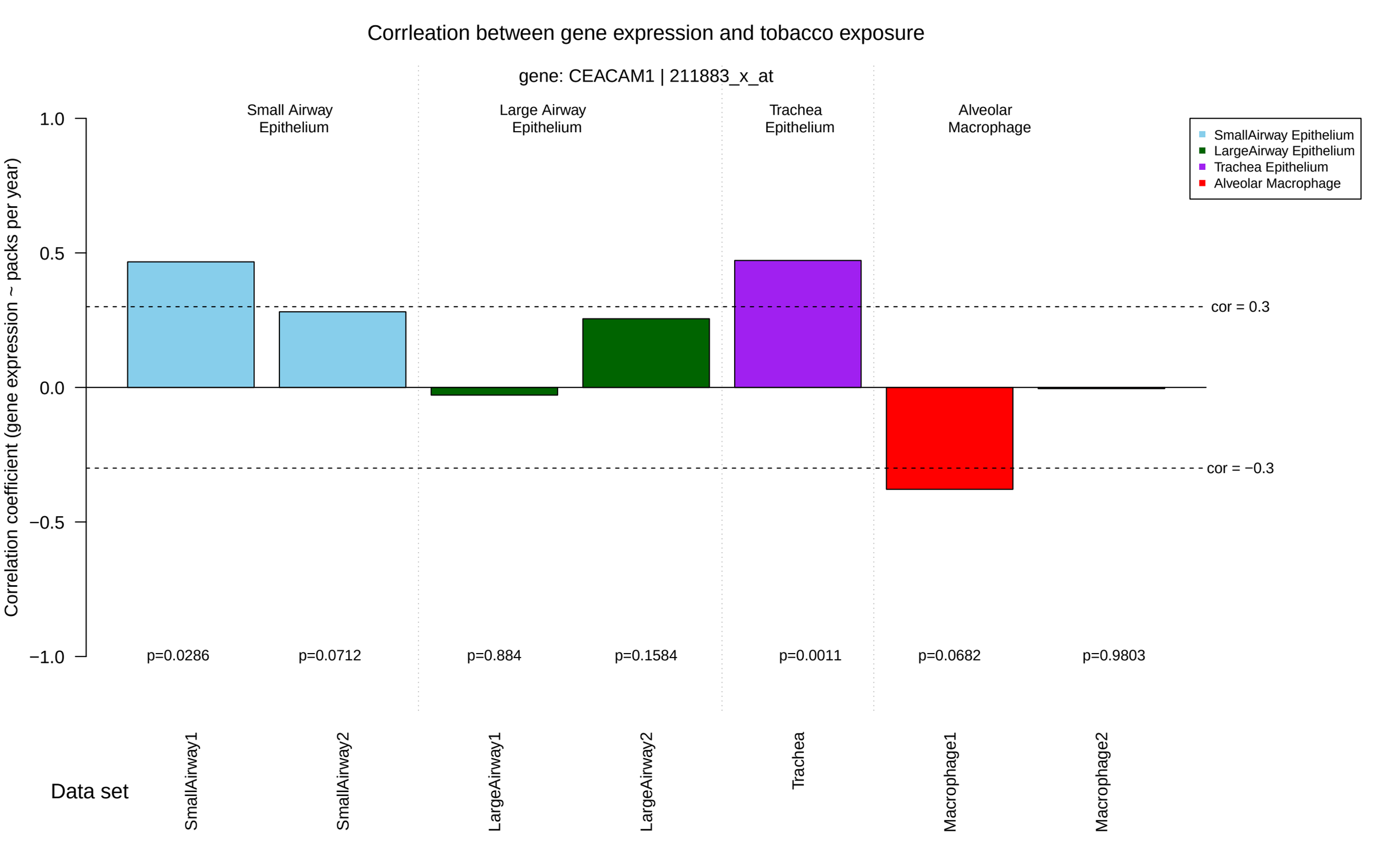

set: LargeAirway1 (p=0.715)
CEACAM1 | 211883_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.002)
CEACAM1 | 211883_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
CEACAM1 | 211883_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5483)
CEACAM1 | 211883_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7847)
CEACAM1 | 211883_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM1 | 211883_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0286
p=0.0712
p=0.884
p=0.1584
p=0.0011
p=0.0682
p=0.9803
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2